University of Rajshahi
(For Affiliated Colleges)
Department of Computer Science and Engineering
B.Sc. Engineering Part-I Even Semester Examination 2021
Course: CSE1211 (Introduction to Digital Electronics)
Syllabus: 2019-2020
Marks: 52.50 Time: 3 Hours

(Answer Six Questions taking at least THREE from each Section.)

Section-A

1.(a)	Convert $(11001010.1101)_2$ to base 16 and $(231.07)_8$ to base 10.	2
(b)	Realize XOR gate using only 4 NAND gates.	1
(c)	Construct a 4-bit odd parity generator circuit using logic gates.	3.75
(d)	Represent $(-142)_{10}$ using 2's complement binary form.	2
2. (a)	What are min terms and max terms?	2
(b)	Design a combinational circuit to convert BCD code to excess-3 code.	3.75
(c)	Convert the following function into sum of product form: (AB+C)(B+CD).	1
(d)	Subtract $(1011)_2$ to $(0100)_2$ using 2's complement method.	2
3. (a)	State and prove DeMorgan's theorem with the help of truth tables.	3
(b)	Simplify $Y = [A(B + C')D]'$.	2.75
(c)	Explain full adder in detail with circuit diagram and truth table.	3
4. (a)	What is Boolean function minimization? What are the advantages of doing minimization?	2
(b)	Simplify the following Boolean function using K-map method: $F(A,B,C,D) = \sum m(0,1,3,7,8,13,14) + d(2,5,9,11)$. Implement the minimized function with logic gates.	6.75

Section-B

5. (a)	What is latch and flip-flop?	2
(b)	Explain master slave J-K flip-flop with timing diagram in detail.	6.75
6. (a)	Distinguish between latch and flip-flop.	2
(b)	Draw a circuit diagram of a D latch and explain its operation.	4
(c)	Draw the circuit diagram of T flip-flop and discuss operation.	2.75
7. (a)	What is sampling and quantization? What are the applications of ADC and DAC?	3
(b)	An R/2R ladder DAC has binary input of 1001. If a HIGH = +5.0 V and a LOW = 0.0 V, what will be equivalent analog output?	2
(c)	Draw the block diagram of ramp ADC and explain its working principle with a suitable example.	3.75
8. (a)	Discuss the operation of a 555 timer IC based monostable multivibrator.	3
(b)	What do you mean by resolution and accuracy for a DAC?	1.75
(c)	Draw the block diagram of ADC and discuss its operation.	4

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engineering Part-I Even Semester Examination 2021
Course: CSE1211 (Introduction to Digital Electronics)
Syllabus: 2020-2021
Marks: 60 Time: 3 Hours

(Answer Six questions taking at least THREE from each Section.)

Section-A

1.(a) Define sign number. What are the uses of 2's complement method in computer system? 4
(b) Perform subtraction using 2's complement method $(77 - 45)_{10}$. 3
(c) Convert $(153.345)_6$ to base 8, base 10 and base 16. 3

2. (a) Show that the Excess-3 code is self-complementing. 4
(b) Convert $(1000\ 0110)_{BCD}$ to decimal, binary and octal. 3
(c) Explain don't care terms with example. Convert $(11001010)_2$ to gray code. 3

3. (a) State and prove DeMorgan's theorem with the help of truth tables. 4
(b) Simplify $Y = [(A' + C)(B + D')]'$. 3
(c) Explain full adder in detail with circuit diagram and truth table. 3

4. (a) Explain prime implicant with example. 3
(b) Simplify the following Boolean function using K-map and realize with basic gates: 7
$F(W, X, Y, Z) = \sum m(0,7,8,9,10,12,) + d(2,5,13)$.

Section-B

5. (a) What are setup and hold times? 3
(b) Explain master slave S-R flip-flop with timing diagram in detail. 7

6. (a) Distinguish between latch and flip-flop. 3
(b) Draw a circuit diagram of a J-K latch and explain its operation. 4
(c) Draw the circuit diagram of T flip-flop and discuss operation. 3

7. (a) Draw the basic DTL gates to implement NAND gate. Explain its operation. 4
(b) Design a TTL NOR gate circuit and briefly explain its operation. 4
(c) State advantages and disadvantages of TTL. 2

8. (a) What is sampling and quantization? What are the applications of ADC and DAC. 4
(b) Draw the block diagram of a successive approximation ADC and explain its working principle with a suitable example. 4
(c) Write the functions of each pin of 555 timer. 2

University of Rajshahi
Dept. of Computer Science and Engineering
B.Sc. Engg. Part-I, Even Semester, Examination 2020
Course Code: CSE 1211 Course Title :Introduction to Digital Electronics

Time: 3 Hours **Full Marks: 52.5**

[N.B. Answer SIX questions taking at least THREE from each Section.]

Section A

1. (a) What are the advantages of encoding a decimal number in BCD as compared to straight binary? What is its disadvantage? 2
   (b) What range of decimal values can be represented by a four-digit octal number? 1.75
   (c) A typical PC uses a 20-bit address code for its memory locations- 3
      i) How many Hex digits are needed to represent a memory address?
      ii) What is the range of addresses?
      iii) What is the total number of memory locations?
   (d) Perform the subtractions $(01001 - 11010)_2$ and $(10010 - 10011)_2$ using 2's complement system. 2

2. (a) Draw the logic diagram and truth table of OR, NOR and XOR gate using NAND gate only. 3
   (b) Simplify the following Boolean expression 3
      i) $X = ABC + A'BC + AC + AC'$
      ii) $A'B\,(D' + CD) + AB + A'BCD$
   (c) Which coding technique is good for error detection? Give example. Convert $(10110)_2 = (?)_{gray}$. 2.75

3. (a) Minimize the following function: $f(a, b, c, d) = \sum a(1, 3, 4, 7, 11) + d(5, 12, 13, 14, 15)$ 3
   (b) Define prime implicant with example. 1.75
   (c) Minimize the following function using Quine-McCluskey method. 4

$$f(a, b, c, d) = \sum_m (0, 1, 2, 4, 6, 8, 12, 14)$$

4. (a) Define the following terms: i) Fan out ii) Noise margin 2.75
   (b) Draw and explain the circuit operation of 2 digit TTL NAND gate. 3
   (c) Design a two input CMOS NAND gate with necessary diagram and truth table. 3

Section B

5. (a) Write down some benefits of clocked flip-flop. Discuss the circuit diagram of the edge triggered SR flip-flop with timing diagram. 4
   (b) Mention some applications of Latch. 1.75
   (c) Briefly explain the operation of master-slave flip-flop using logic and timing diagram. 3

6. (a) Draw the internal functional diagram of a 555 timer and explain its basic operation. What are the applications of 555 timer? 4.75
   (b) Draw the pin configuration of a 555 timer IC and explain the function of each pin. 4

7. (a) Define resolution or step size of a D/A converter. For a DAC, if step size is 0.1 V, then what will be the output voltage for a digital input 0001. 2.75
   (b) What is the advantage of R/2R ladder DAC over weighted registers DAC? Draw and explain the basic operation of R/2R ladder DAC. 3
   (c) What is the main advantage of a SAC over digital ramp ADC? An 8-bit SAC has resolution of 20 mV. What will its digital output be for an analog input of 2.17 V? Illustrate with diagram. 3

8. Write short notes on the following (any three): 8.75
   (a) ECL NOR/OR gate
   (b) Shift register
   (c) Digital Ramp ADC
   (d) 3 bit counter

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg.(CSE) 1st Year Even Semester 2019
Course: CSE1211 (Introduction to Digital Electronics)
Time: 3 Hrs. Full Marks: 52.5
[N.B. Answer SIX questions taking at least THREE from each Section.]

Section A

1. (a)	Perform subtraction using 2's complement method $(85\text{-}47)_{10}$.	3
(b)	convert $(541.203)_6$ to base 5, base 8 and base 10.	3
(c)	Write the binary code for the gray code 10110111.	0.75
(d)	Represent the decimal number 28 to excess-3 and BCD code.	2
2. (a)	Given the boolean function K = AB+A'B'+B'C Then i) implement it with AND, OR and NOT gates ii) implement it with only OR and NOT gates.	3
(b)	Show that the dual of the exclusive-OR is equal to its complement.	3.75
	Simplify the following expression $z = \bar{A}B\bar{C} + AB\bar{C} + B\bar{C}D$	
(c)	What are don't care terms? Explain with example.	2
3. (a)	What is parity bit? Design 3-bit odd parity generator and checker.	6
(b)	Implement a full adder using two half adder.	2.75
4. (a)	Design a logic circuit whose output is HIGH only when a majority of inputs A, B and C are LOW.	2.75
(b)	Simplify the following Boolean function using K-map and realize with basic gates: $F(A, B, C, D) = \sum m(1, 5, 6, 12, 13, 14) + \sum d(2, 4)$.	6

Section B

5(a)	Show the logic diagram of a clocked RS flipflop with four NAND gates.	3
(b)	With the help of timing diagram, explain the operation of positive edge triggered JK FF.	3
c)	What are the advantages of a master-slave JK FF? Mention some limitations of SR FF.	2.75
6(a)	What is the basic difference between latch and flip-flop?	1
(b)	Define setup time, propagation delay and hold time for flip-flop using proper timing diagram.	2.75
(c)	Draw the circuit diagrams and discuss the operation of D flip-flop and T flip-flop.	5
7.(a)	Draw the TTL NAND gate circuit and explain its operation.	3
(b)	What is current sourcing and current sinking action?	1.75
(c)	Draw and explain a CMOS NOR gate circuit.	4
8.(a)	Discuss the operation of a 555 timer IC based monostable multi-vibrator.	3
(b)	A certain binary-weighted-input DAC has a binary input of 1101. If a HIGH = +3.0 V and a LOW = 0 V, what is V_{out}?	1.75
(c)	Draw the block diagram of a successive-approximation ADC and discuss its basic operation with a flow-chart.	4

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg.(CSE) 1st Year Even Semester 2018
Course: CSE1211 (Introduction to Digital Electronics)
Time: 3 Hrs. Full Marks: 52.5
[N.B. Answer SIX questions taking at least THREE from each Section.]

Section A

1. (a) Find $(45)_{10} - (83)_{10}$ using two's complement format with 8-bit numbers. Then convert your result back to decimal. 3
   (b) Add $(65)_{10} + (72)_{10}$ using 8-bit sign-magnitude format for the numbers. Convert your result to decimal. Is your answer correct? Why or why not? 5
   (c) Write the gray code for $(1011)_2$. 0.75

2.(a) Convert $(1000\ 0110)_{BCD}$ to decimal, binary & octal. 3
   (b) Simplify Z = A' C (A' B D)' + A' B C' D' + A B' C using Boolean algebra. 2.75
   (c) State & prove De Morgan's theorems with the help of truth tables. 3

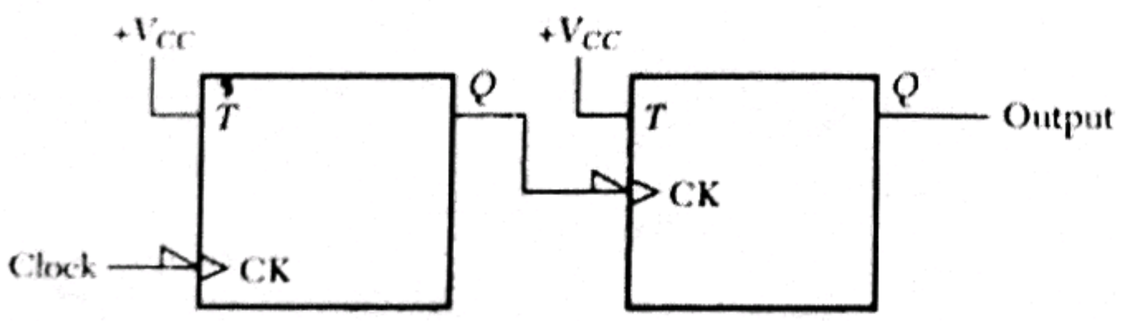
3. (a) Design and explain a full adder in detail with circuit diagram and truth table. 4
   (b) Design a combinational logic circuit to compare two 2-bit binary numbers A and B and to generate the outputs A<B, A=B and A>B. Is there a way to derive the third output from the first two outputs? 4.75

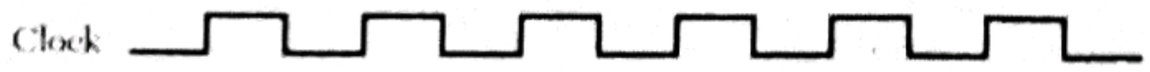
4. (a) Simplify the following Boolean expression using Quine-McCluskey technique $f(A,B,C,D)= \Sigma m(0,1,3,7,8,9,11,15)$. 6
   (b) List out the advantages and disadvantages of Quine-Mc Cluskey Method. 2.75

Section B

5(a) What is latch and flip-flop? 2
   (b) Explain the master slave S-R flip-flop with timing diagram in detail. 6.75

6(a) Draw a circuit diagram of a J-K latch and explain its operation. 4
   (b) Plot the output waveforms referenced to the clock signal assuming the initial contents of all FFs is Q = 0. Assume all FFs are edge triggered. 4.75





7.(a) State advantages and disadvantages of TTL. 2
   (b) Draw the circuit diagram of DTL-NAND gate. 3
   (c) Design a NOT gate using MOSFET. 3.75

8.(a) Draw the block diagram of a system to interface a computer to the analog world so that the computer can monitor and control a physical variable and explain the functions of each block. 4
   (b) What do you mean by resolution and accuracy for a DAC? 1.75
   (c) Draw the block diagram of an ADC and discuss its basic operation. 3

University of Rajshahi
Department of Computer Science and Engineering
B.Sc.(Engg.), Part-1, Even Semester Examination-2017
Course: CSE1211 (Introduction to Digital Electronics)
Time: 3 Hrs. Full Marks: 52.5
[N.B. Answer any **Three** questions from each Section.]

Section A

1. (a) Convert $(FBA.BC)_{16}$ to decimal and $(486.524)_{10}$ to binary. 3
   (b) Convert $(465.32)_8$ to decimal and $(11010101.1101)_2$ to Decimal. 3
   (c) Write the Excess-3 code for $(53)_{10}$. Attach an even parity bit with the generated code. 2.75

2. (a) Perform the following operations of the singed binary numbers using 2's complement method: 3
   (i) 00001100 – 11110111 (ii) 10001000 – 11100010
   (b) Show that both NAND gate and NOR gate can be used as universal gate. 3
   (c) Multiply the signed binary numbers: 01010011 (multiplicand) and 11000101 (multiplier). 2.75

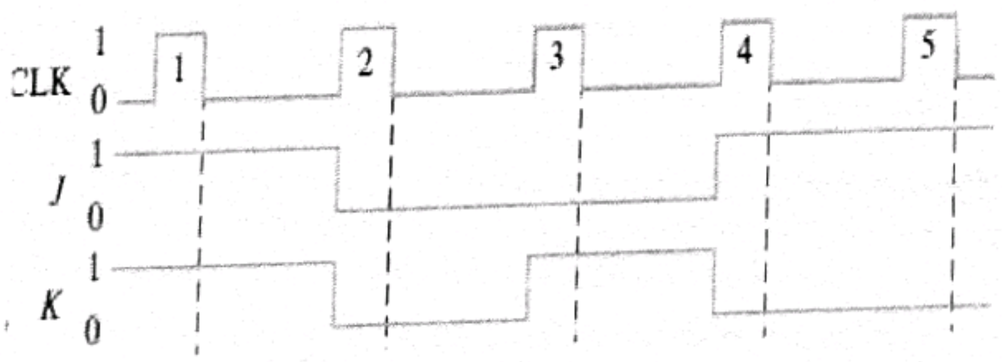
3. (a) Explain half adder and full adder in detail with circuit diagram and truth table. 6
   (b) What is DeMorgan's theorem? Explain with truth table. 2.75

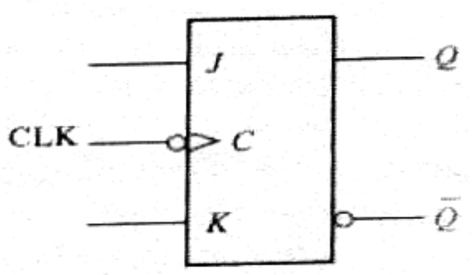
4. (a) Explain the formation of AND, OR and NOT gate using NAND gate only. 2.75
   (b) Simplify the following Boolean function using K-map and realize with basic gates: 6
   $F = (A, B, C, D) = \sum m\,(0, 1, 3, 5, 7, 8, 9, 13) + \sum d\,(10, 11, 15)$.

Section B

5. (a) What are the functions of transducer and actuator? 2
   (b) Describe the basic operation of the digital-ramp ADC. 3.75
   (c) A 5-bit DAC produces V_{OUT}=0.2 V for a digital input of 00001. Find the value of V_{OUT} for an input of 11111. What would be the resolution? Describe the staircase signal out. 3

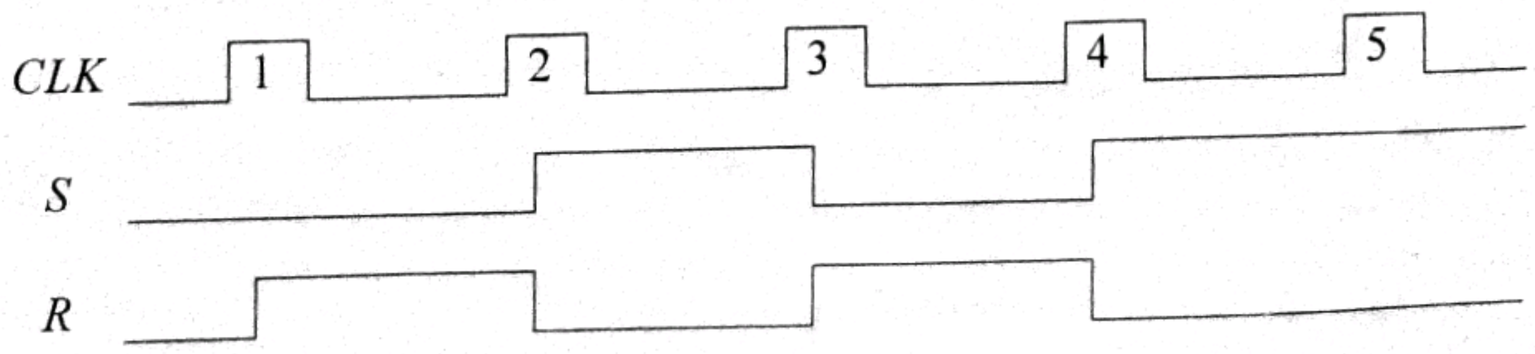
6(a) Draw a circuit diagram of a D latch and explain its operation. 4
   (b) The waveforms in Figure are applied to the J, K, and clock inputs as indicated. Determine the Q output, assuming that the flip-flop is initially RESET. Also explain about the states of the Q output according to the waveforms. 4.75





7.(a) Draw the CMOS NAND gate circuit and explain its operation. 3
   (b) Draw the TTL NOR gate circuit and explain its operation. 4
   (c) Define the terms: (i) Fan-Out (ii) Propagation Delay and (c) Noise Immunity. 1.75

8. (a) Draw the internal functional diagram of a 555 timer and explain the basic operation. 3.75
   (b) For a certain astable multivibrator t_H = 15 ms and T = 20 ms. What is the duty cycle of the output? 2
   (c) Determine the waveforms for both outputs of a positive edge triggered S-R flip-flop for the following S, R and CLK input. Assume that the flip-flop is initially RESET. 3



University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg.(CSE) 1st Year EVEN Semester 2016
Course: CSE1211 (Introduction to Digital Electronics)

Time: 3 Hrs. **Full Marks: 52.5**

[N.B. Answer SIX questions taking at least THREE from each Section.]

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Part A

1. (a)	Convert $(1001010.1101001)_2$ to base 16 and $(231.07)_8$ to base 10.	2
(b)	Realize XOR gate using only 4 NAND gates.	1
(c)	Construct a 4-bit odd parity generator circuit using logic gates.	3.75
(d)	Represent $(-200)_{10}$ using 2's complement binary form.	2
2. (a)	What are min terms and max terms?	2
(b)	Design a combinational circuit to convert BCD code to excess-3 code.	3.75
(c)	Convert the following function into sum of product form: (AB+C)(B+C'D)	1
(d)	Subtract $(1011)_2$ from $(0100)_2$ using 2's complement method.	2
3. (a)	A small process-control computer uses hexadecimal codes to represent its 16 bit memory address. Answer the following: (i) How many Hex digits are required? (ii) Write the range of addresses in binary and in Hex. (iii) In Hex, how many memory locations are there?	3
(b)	Design a logic circuit that will allow a signal to pass to the output only when control inputs B and C are both HIGH; otherwise the output will stay LOW.	1
(c)	You have to design a logic circuit that controls an elevator door in a two-story building. Consider a logic signal say O that indicates when the elevator is moving (O=1) or stopped (O=0). A_1 and A_2 are floor indicator signals that are normally LOW, and they go HIGH only when the elevator is positioned at the level of that particular floor. Draw the truth table for open and from it, write the logic expression and design the logic circuit. (For example, when elevator is lined up with the first floor, A_1 =1 and A_2 =0. The circuit output is the open signal, which is normally LOW and will go HIGH when the elevator door is to be opened)	4.75
4.	Simplify the following Boolean function using Quine-McClusky method F = (A, B, C, D, E)= $\sum$ m (0, 1, 3, 7, 8, 13, 14, 21, 26) + $\sum$ d (2, 5, 9, 11, 17, 24).	8.75

Part B

5(a)	Explain the master slave J-K flip-flop with timming diagram in detail.	7
(b)	What is setup and hold times?	1.75
6(a)	Define the following terms: (i) Fan in and fan out (ii) Noise immunity	2.75
(b)	Design a TTL NAND gate circuit and beriefly explain its operation.	3
(c)	Design the circuit diagram of DTL-NAND gate and beriefly explain its operation.	3
7(a)	What is sampling and quantization? What are the applications of ADC and DAC?	3
(b)	An R/2R ladder DAC has binary input of 1001. If a HIGH = +5.0 V and a LOW = 0.0V, what will be equivalent analog output?	2
(c)	Draw the block diagram of a successive approximation ADC and explain its working principal with a suitable example.	3.75
8(a)	Write the funtions of each pin of 555 timer.	2.75
(b)	Design an astablemultivibrator using 555 timer and explain its operation.	4
(c)	An Astable 555 multivibratoris constructed using the following components, R1 = 1kΩ, R2 = 2kΩ and capacitor C = 10μF. Calculate the output frequency from the multivibrator and the duty cycle of the output waveform.	2

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg. (CSE) Part-1 Even Semester Exam-2014
Course: CSE 1211 (Introduction to Digital Electronics)
Full Marks:52.5 **Time: 3 Hours**

[N.B. Answer SIX questions taking THREE from each part]

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University of Rajshahi

Part-A

1. (a) Do the following conversions: 4
(i) $(10101011.1101)_2=(?)_{10}$ (ii) $(20345.125)_{10}=(?)_2$ (iii) $(80914.25)_{10}=(?)_8$
(iv) $(3AE8F.2D)_{16}=(?)_8$ (v) $(1011001110)_2=(?)_4$
(b) Add $(110111)_2$ with $(100111)_2$. Subtract $(100110)_2$ from $(110011)_2$ using 2's complement method. 3
(c) Represent $(-17)_{10}$ in sign magnitude, 1's complement and 2's complement representation. 1.75

2. (a) Write the procedure to convert a binary code to gray code with example. 4
(b) With the help of example explain excess-3 code. 3
(c) Write the BCD code for $(9248)_{10}$. 0.75
(d) How can you easily generate 3 bit gray code . 1

3. (a) Define and draw the truth table of a 3-input X-OR gate. 2.75
(b) Show that NAND gate can be used as universal gate. Simplify the following using De-Morgan's theorem 3
$\left(\overline{A+\overline{B}}\right)\left(\overline{\overline{A}+B}\right)$
(c) Simplify the following Boolean expressions to a minimum number of literals: 3
(i) $ABC+A'B+ABC'+AC$ (ii) $A'B(D'+CD)+B(A+A'CD)$

4. Simplify the following logic function using Quine-McCluskey technique 8.75
$f(A,B,C,D) = \sum m(0,1,3,7,8,9,11,15)$

Part-B

5. (a) Write the difference between asynchronous and synchronous system. 1.75
(b) Explain the operation of a positive-edge triggered SR flip-flop by timing diagram. 4
(c) Discuss how an SR latch is converted into D-latch. 3

6. (a) Explain the operation of a JK flip-flop using timing diagram. 3.75
(b) Mention the limitations of a JK flip-flop. 2
(c) Describe how a D-latch operates differently from an edge-triggered D flip-flop. 3

7. (a) Discuss the operation of an 8-bit DAC using op-am summing amplifier with binary weighted registers. 3
(b) Define R/2R ladder DAC with basic circuit diagram. Write the benefits of R/2R circuit. 3
(c) Briefly discuss the operation of the digital ramp ADC. 2.75

8. (a) Draw the block diagram of a 555 timer and explain about its each pin. 3
(b) Design an astable multi-vibrator using 555 timer and explain its operation. 5.75

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University of Rajshahi
Department of Computer Science and Engineering
B.Sc. (Engg.) Examination-2013, Year-I, Semester-II
Course: CSE-1211 (Introduction to Digital Electronics)
Full Marks-52.5 Time: 3 hours

[N.B. Answer any six questions taking THREE from each of the groups]

Part-A

1. a) Convert 1001.1001 into decimal. Perform the 2's complement operation on $(10100)_2$. — 3
   b) What are the advantages of Gray code? Convert the number $(0101)_2$ to ets Gray code equivalent. — 3
   c) Subtract $(1011)_2$ from $(0101)_2$ using 1's and 2's complement method. — 2.75

2. a) Define and draw the truth table of a 3-input X-NOR gate. — 2.75
   b) Show that NOR gate can be used as universal gate. Simplify the following using De Morgan's theorem $(\overline{M+\bar{N}})(\overline{\bar{M}+N})$. — 3
   c) Define Active LOW and Active HIGH with necessary diagram. — 3

3. a) Minimize the Boolean function f(a, b, c, d)=$\sum_M(1,3,6,8,10)$. — 4
   b) Minimize the following function using Quine-Mc Cluskey method (up to level one): f(a, b, c, d)= $\sum_M(0,1,2,4,6,8,12,14)$. — 4.75

4. a) What do you mean by fan in and fan out? — 2.75
   b) Draw the circuit diagram of DTL-NAND gate. — 3
   c) Design a NOT gate using MOSFET. — 3

Part-B

5. a) Explain the operation of NAND gate latch with its logic circuit and typical waveform. — 4
   b) What is the difference between asynchronous and synchronous system. — 2.75
   c) Mention some application of FF. — 2

6. a) How does the binary counter work as a frequency divider? — 2
   b) What is the frequency of the output of 8th FF when the input clock frequency is 512 KHz? — 1.75
   c) Design a full adder circuit. — 5

7. a) Define Multiplexer. How can you design an eight-input MUX by using two 4-input MUX. — 3
   b) What is the difference between DeMUX and decoder? — 1
   c) Design a decimal weighted table in binary equivalent for a three-digit BCD number. Using the table convert 100100110101(BCD) to binary. — 4.75

8. Write short notes (any three) from the following: — 8.75
   a) Successive Approximation ADC
   b) Analog to Digital Conversion.
   c) TTL and METL.
   d) Shift Register.

University of Rajshahi

Dept. Of Computer Science and Engineering

B. Sc. Engg.(CSE) 1st Year Even Semester Examination 2012

Course: CSE 1211 (Introduction to Digital Electronics)

Full Mark: 52.5 Duration: 4 hours

Answer 6 questions taking at least 3 from each part

Part -A

1.	a)	What is coding? Discuss how parity bits can be used to detect error in bits.	2
	b)	Convert 146_{10} to octal, then from octal to binary.	1.75
	c)	Mention some advantages of using BCD compared to other coding.	1.5
	d)	Perform the 2's compliment operation on $(10000)_2$	1.5
	e)	Add the (1010+1011) in binary. Check your result by doing the addition in decimal.	2
2.	a)	What do you mean by minterm and maxterm?	1
	b)	Simplify the following Boolean expressions to a minimum number of literals: i) $ABC + A'B + ABC' + AC$ ii) $A'B(D' + CD) + B(A + A'CD)$	2
	c)	Minimize the following switching function using Quine-McCluskey method. $f(w,x,y,z) = \sum_m (0,1,2,4,5,6,8,9,12,13,14)$	5.75
3.	a)	Which gates are called Universal gate? Why?	2.75
	b)	Minimize the Boolean function $f(x_1,x_2,x_3,x_4) = \sum_m (0,5,7,8,9,10,11,14,15)$ using k-map.	4
	c)	State and prove De-Morgan's law.	2
4.	a)	Draw and explain the circuit operation of a 2-input TTL NOR gate.	3
	b)	Draw and explain the circuit operation of a 2-input CMOS NOR gate.	4
	c)	Write a short note on the ECL family.	1.75

Part -B

5.	a)	Explain the operation of a positive-edge triggered SR flip-flop by timing diagram.	4
	b)	How can you convert an SR latch into a D-latch?	3
	c)	Draw the circuit diagram of a master/slave JK flip-flop.	1.75
6.	a)	With the help of timing diagram, explain the operation of a clocked JK flip-flop.	3
	b)	What are the limitations of a JK FF?	2
	c)	Describe how a D flip-flop operates differently from an edge triggered D flip-flop.	3.75
7.	a)	Draw and explain the operation of 555 timer as astable multivibrator.	4
	b)	Briefly discuss a simple DAC using Op-Amp.	3
	c)	What is timer? Mention some applications of timer circuit.	1.75
8.	a)	What is the function of a transducer and actuator?	1.75
	b)	Draw the circuit diagram of R-2R ladder DA Converter.	1.5
	c)	Which factor affect DA conversion accuracy?	1.5
	d)	Define full scale error and settling time.	2
	e)	Briefly discuss the operation of the digital ramp ADC. Write its applications.	2

EVEN

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg.(CSE) 1st Year 2nd Semester 2011
Course: CSE 1211 (Digital Electronics-II)
Time: 4 Hrs. **Full Marks: 52.5**
[N.B. Answer SIX questions taking at least THREE from each part.]

PART-A

1. a) Draw the diagram of a NAND gate latch and show that it has two possible resting states when SET=RESET=1. 2
b) With the help of timing diagram, explain the operation of positive-edge triggered J-K flip-flop. 3
c) What are hold time and setup time? 1.75
d) Discuss the potential timing problem in FF circuits with necessary timing diagram. 2

2. a) What is shift register? Discuss with timing diagram the operation of a four-bit shift register using J-K flip-flop. 2.5
b) Show that J-K flip-flop can be used as a binary counter. 2.5
c) What is free running multi-vibrator? Discuss the circuit operation of a 555 timer IC as a clock generator circuit. 3.75

3. a) What is D latch? Describe how a D latch operates differently from an edge-triggered D flip-flop. 3
b) What is the impact of propagation delay between the transitions of a flip-flop? 1
c) Define shift-register. Construct two 3-bits shift registers X and Y, and show how the bits stored in register X are transferred to register Y with a shift pulse. Explain. 4.75

4. a) What is priority encoder? Discuss the logic circuit for an 8-to-3 encoder with diagram. 3
b) Draw the circuit for keyboard entry of 2-digit number into storage register. 2.75
c) Discuss the circuit of a four-input multiplexer with function table and diagram. 3

PART-B

5. a) Differentiate between asynchronous counter and synchronous counter. 2
b) Show that how the 74LS293 IC wired as a MOD-16 counter with a 10KHz clock input. Determine the frequency at Q_2. 2
c) What is the difference between the counting sequences of an up-counter and a down-counter? 1.75
d) Describe the circuit operation of a MOD-8 down-counter with timing diagram. 3

6. a) Draw the logic diagram for a BCD-to-Decimal decoder. Give its logic symbol and truth table. Explain how it can be combined with a counter to provide timing and sequencing operations. 6
b) Can more than one decoder output be activated at one time? Explain. 2.75

7. a) Discuss the operation of a 8-bit DAC using an op-amp summing amplifier with binary weighted registers. 2
b) What is R/2R ladder? What are the benefits of R/2R circuit? Draw the basic circuit of R/2R ladder DAC. 3
c) How a DAC can be used to control the amplitude of an analog signal? Discuss with diagram. 2
d) Draw the circuit diagram of digital RAMP ADC. 1.75

8. Write short notes on the followings (any three): 8.75
   a) Analog to digital conversion.
   b) Master/Slave flip-flops.
   c) BCD-to-7 segment decoder/driver.
   d) 1-to-8 line De-multiplexer.